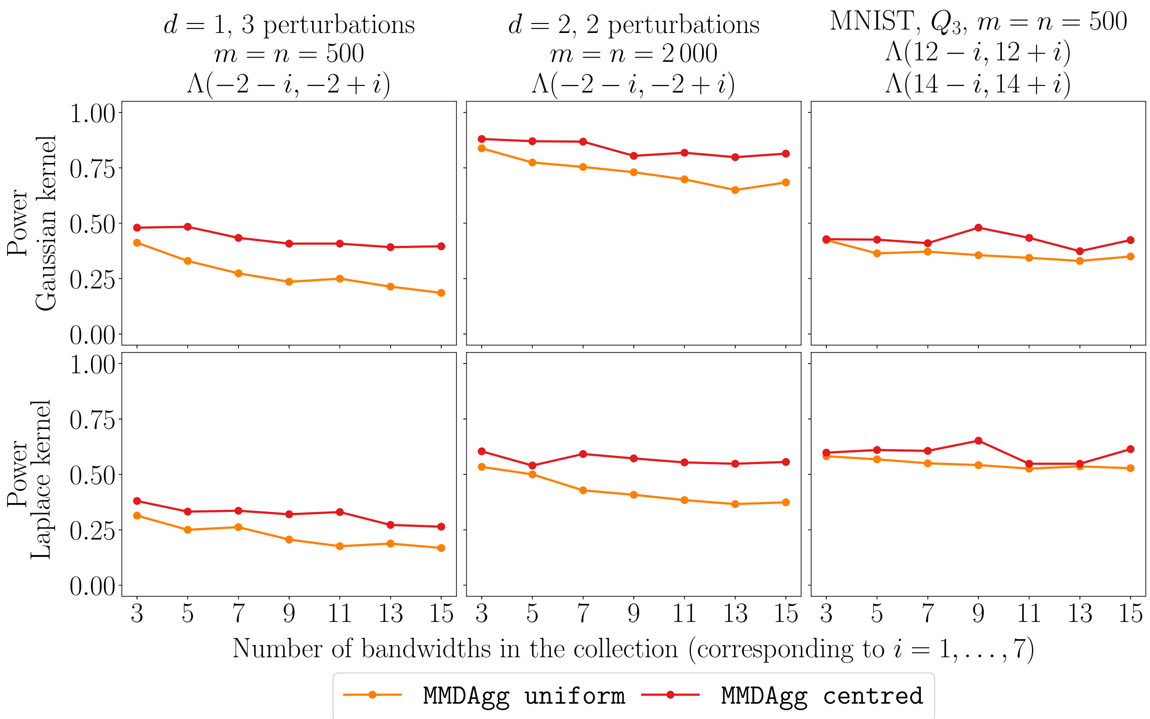

Figure 7: Power experiments varying the size of the collections of bandwidths using perturbed uniform d-dimensional distributions and the MNIST dataset with a wild bootstrap.

the Gaussian and Laplace kernels, we use the collections of bandwidths $\Lambda(12 - i, 12 + i)$ and $\Lambda(14 - i, 14 + i)$ for $i = 1, \ldots, 7$, respectively. Those collections are centred as those corresponding to the middle columns of Figures 3 to 5 (for the case $i = 2$), so we expect the bandwidth in the centre of each collection to be a well-calibrated one. For this reason, it makes sense to consider only `MMDAgg uniform` and `MMDAgg centred` in those experiments.

In all the settings considered in Figure 7, we observe only a very small decrease in power when considering a wider collection of bandwidths for `MMDAgg centred`. This is due to the fact that even though we consider more bandwidths, we still put the highest weight on the well-calibrated one in the centre of the collection. Nonetheless, the fact that almost no power is lost when considering more bandwidths for `MMDAgg centred` is a great feature of our test, which is only possible due to the way we perform the level correction for MMDAgg. For the MNIST dataset, we observe a slight increase in power for `MMDAgg centred` with the collections of nine bandwidths for both kernels. This could indicate that, as suggested in Section 5.6, the bandwidths in the centre of the collections are well-calibrated to distinguish $\mathcal{P}$ from Q_4 but are not necessarily the best choice to distinguish $\mathcal{P}$ from Q_3.

Remarkably, the power for `MMDAgg uniform`, which puts equal weights on all the bandwidths, decays relatively slowly and this test does not use the information that the bandwidth in the centre of the collection is a well-calibrated one. So, we expect similar results for any collections of the same sizes which include this bandwidth but not necessarily in the centre of the collection. This means that, in practice, without any prior knowledge, one

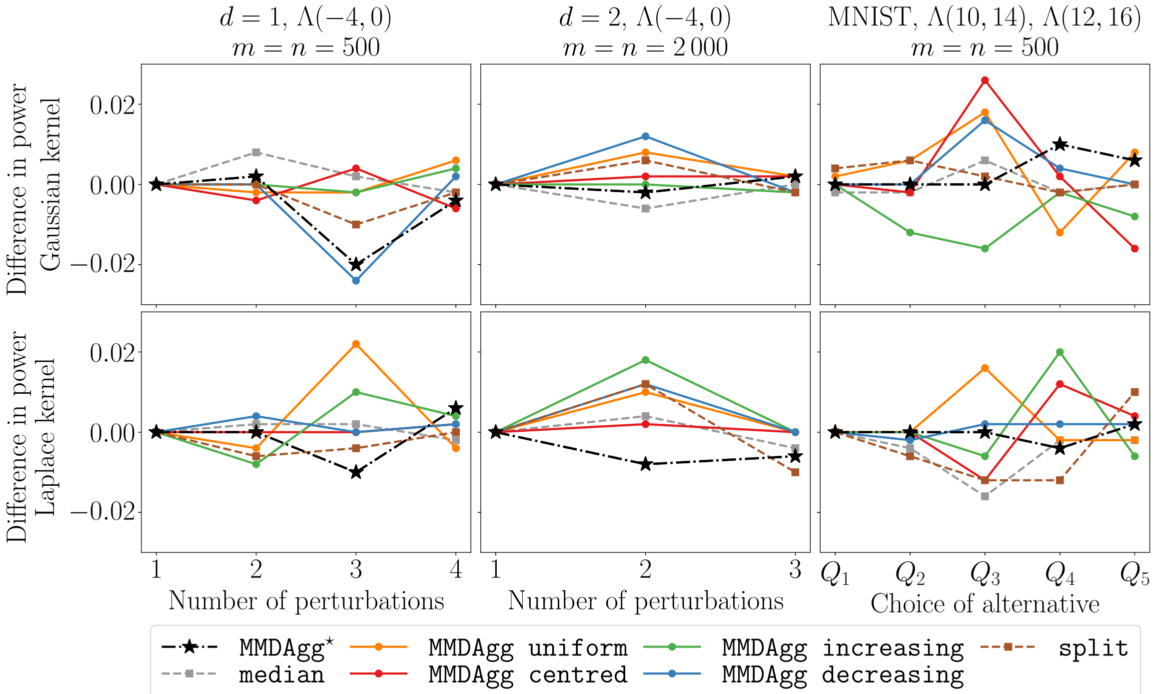

Figure 8: Power experiments considering the difference between using a wild bootstrap or permutations on perturbed uniform d-dimensional distributions and on the MNIST dataset.

can use uniform weights with a relatively wide collection of bandwidths without incurring a considerable loss in power.

A.3 Power experiments: comparing wild bootstrap and permutations

We consider the settings of the experiments presented in Figures 3 to 5 on synthetic and real-world data using the Gaussian and Laplace kernels. We run the same experiments using permutations instead of a wild bootstrap for one collection of bandwidths for each of the different settings. We then consider the power obtained using a wild bootstrap minus the one obtained using permutations, and plot this difference in Figure 8.

The absolute difference in power between using a wild bootstrap or permutations is minimal, it is at most roughly 0.02 and is even considerably smaller in most cases. Furthermore, the difference overall does not seem to be biased towards using either of the two procedures. Since there is no significant difference in power, we suggest using a wild bootstrap when the sample sizes are the same since our implementation of it runs slightly faster in practice. Of course, when the sample sizes are different, one must use permutations.

A.4 Power experiments: using unbalanced sample sizes

In Figure 9, we consider fixing the sample size m and increasing the size n of the other sample, we use permutations since we work with different sample sizes. We consider the

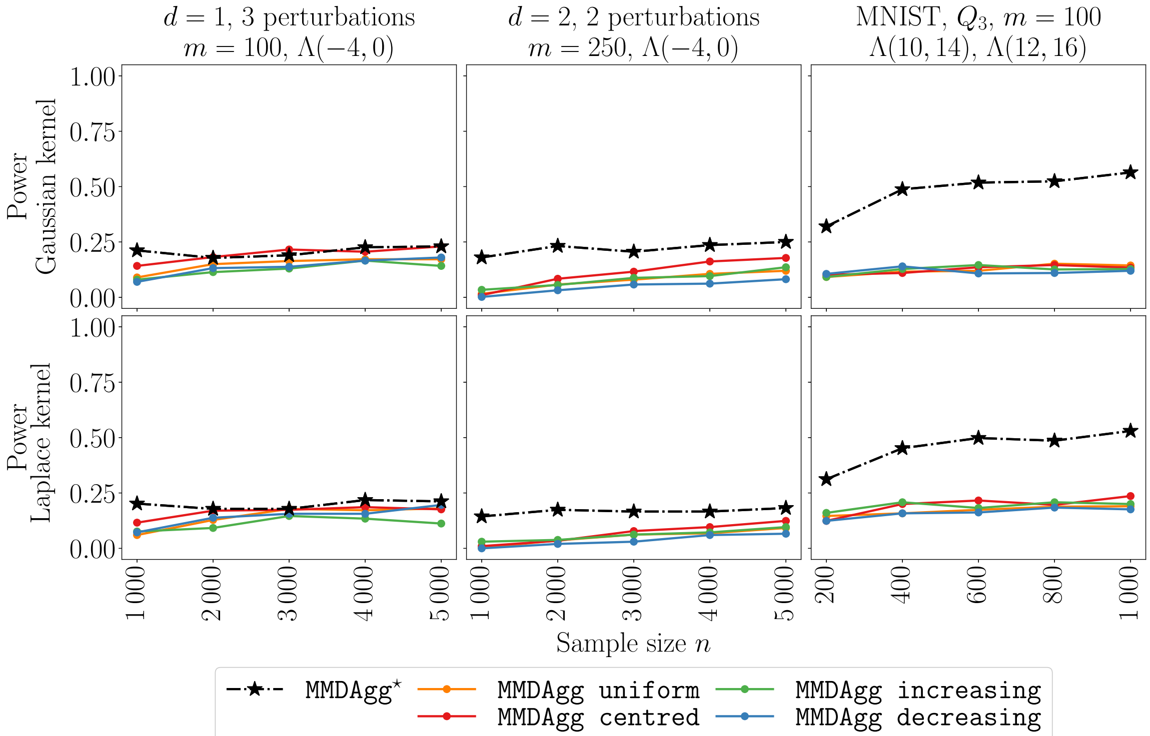

Figure 9: Power experiments with different sample sizes $m \neq n$ on perturbed uniform d-dimensional distributions and on the MNIST dataset using permutations.

settings of Figures 3 to 5 with three and two perturbations for the uniform distributions in one and two dimensions, respectively. For the MNIST dataset, we use the set of images of all digits $\mathcal{P}$ against the set of images Q_3 which does not include the digits 4, 6 and 8.

We observe the same patterns across the six experiments presented in Figure 9. When fixing one of the sample sizes to be small (100 or 250), we cannot achieve power higher than 0.25 (except for `MMDAgg`* on the MNIST data, which as in Figure 5 performs much better than all other tests) by increasing the size of the other sample to be very large (up to 5000). Indeed, we observe that a plateau is reached where considering an even larger sample size does not result in higher power. In some sense, all the information provided by the small sample has already been extracted and using more points for the other sample has almost no effect. As shown in Figures 3 to 5, we can obtain significantly higher power in all of those settings using samples of sizes $m = n = 500$ or $m = n = 2000$. Having access to even more samples overall (5100 instead of 1000) but in such an unbalanced way results in very low power. This shows the importance of having, if possible, balanced datasets with sample sizes of the same order.

A.5 Power experiments: increasing sample sizes for the `ost` test

In Figure 10, we report the results of Figures 3 to 5 for `MMDAgg uniform` and `ost` which are run using the same collections of bandwidths. As previously mentioned, the `ost` test